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A CFD Method for Optimal Air-Intake Design

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Abstract An adjoint optimization method, based on the solution of an inverse problem, is extended to the case of viscous flows. Given a certain performance functional, it is required to find its extremum with respect to a flow variable distribution on the domain boundary, e.g. the pressure. The adjoint formulation delivers the functional gradient with respect to such a flow variable distribution, and a descent method can be used for optimization. The flow constraints are easily imposed in the parameterization of the distributed control, and therefore those problems with several strict constraints on the flow solution can be solved very efficiently. The methodology is here applied to the optimal design of air-intakes and diffusers.

Keywords Optimal Design · Inverse Problems · Adjoint method

1 Introduction

Aerodynamic design can be assisted in two essentially different ways. One, the classic approach, is based on the inverse problem solution, the other, more recent, relies on numerical optimization. In the inverse problem it is usually required to determine unknowns that are given in the natural, i.e., direct position of the problem. For example, a typical inverse problem is to find the airfoil geometry, given the flight speed and the pressure distribution on its surface. In the classical works of Mangler [1] and Lighthill [2] the airfoil inverse problem was solved in the framework of potential flows and using conformal mapping. Further developments of this solution method are extensively accounted in [3] and are related to the introduction of viscous models for laminar and turbulent flows and to the solvability conditions of the problem. Other examples of inverse problem solution methods are found in the volume AGARD-R-780, 1990. In addition, Polito's [4] approach, relative to the spectral solution of the inverse problem for airfoils, and that of Bauer et al.[5] for shockless airfoils, should be mentioned. One drawback of inverse problems is that they may be ill posed: requiring certain wall pressure distributions on airfoils, results in open or self-intersecting profiles. Lighthill discovered the solvability conditions that should be respected by pressure distributions within an incompressible potential flow model, whereas the solvability conditions for compressible flows were investigated in [6] and references therein. The appeal of inverse problem solution for aerodynamic design declined as powerful computers and sound numerical methods that allow numerical optimization of aerodynamic components, became available. Once a functional that defines the relative merit or cost of a certain solution is defined, a numerical optimization algorithm can be as simple as i) computing the functional gradient relative to the controls by divided differences, ii) marching toward the functional extremum using gradient information. If the number of parameters that have to be optimized is in fact not very small, finding the gradient requires a huge amount of computational time. The computation of each gradient component needs a flow evaluation, making optimizations using the Euler or Navier-Stokes models unfeasible. Greater computational efficiency is obtained by using the adjoint method, see [7] [8] [9], to com-

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pute the functional gradient. Evaluation of the gradient requires one adjoint calculation and one flow calculation, regardless of the number of design variables. This approach has opened the possibility of optimizing tridimensional compressible viscous flows over wing-body configurations at high Reynolds numbers[10]. The advantages of numerical optimization over the inverse problem can be summarized in that numerical optimization allows the maximization or minimization of global quantities such as lift or drag in the presence of constraints, whereas for inverse problems the design is limited to the pressure distribution selection on the boundary, which is given on the basis of designer experience, and therefore somewhat arbitrarily. In addition, no control of the final geometry is possible. These weaknesses have been overcome by extending the adjoint optimization method to inverse problems [11]. To formulate a shape optimization problem we need a functional $\mathcal{F}$ to be either minimized or maximized. We have $\mathcal{F} = \mathcal{F}\left[U(\Gamma), \Gamma\right]$ where U is the flow variables vector and Γ represents the geometry. In the adjoint method such an extremum problem is solved using a variational technique and introducing a Lagrange multiplier vector Λ dual of the flow variables vector. Using the Lagrange multipliers we are able to write $\delta\mathcal{F}$ for $\Gamma \leftarrow \Gamma + \delta\Gamma$, at the cost of solving a system of PDEs for the Lagrange multipliers, which is the mathematical adjoint of the governing equations. Once the gradient is known, the initial geometry is perturbed accordingly and the procedure is started all over again until a convergence criterion is satisfied. The adjoint method can be adapted to an inverse problem formulation. Let $p(s)$ be the flow quantity we prescribe on the flow-field boundary, where for example p is the pressure and s is the curvilinear coordinate along the boundary. We define a cost/merit functional in much the same way as before: $\mathcal{F} = \mathcal{F}\left[U(p), p\right]$. It should be noted that the control is now the pressure distribution on the boundary, whereas in usual adjoint methods the control is the boundary shape. The derivation of the adjoint follows the same steps as in the shape optimization case, to finally obtain $\delta\mathcal{F}$ for $p(s) \leftarrow p(s) + \delta p(s)$. The pressure distribution is then altered according to the gradient information until the extremum is eventually reached. In this formulation the boundary shape results from the solution of the inverse problem corresponding to each optimization step. In this respect the optimization of an inverse problem can be considered as a *flow design optimization* as opposed to the optimization of a direct problem, which is known as *shape design optimization*. The flow design optimization offers a very simple way of implementing flow constraints, as they can be directly included in the parameterization of the control. Inevitably, in the design process it is necessary to focus on a given model to account for the physically relevant phenomena which affect performance. Yet, it is mandatory to include results obtained by more sophisticated models or other disciplines, in the selected model. These results usually take the form of constraints on the governing equation variables, and not on the geometry. In [11] the method, based on the Euler equation of motion, is compared with a *classical* adjoint formulation. where the controls was the position of the

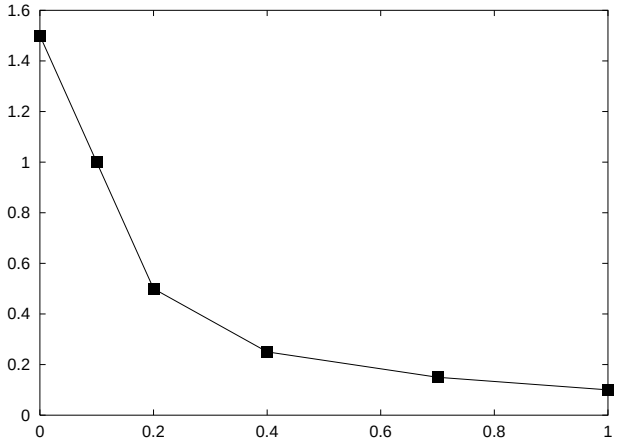


Fig. 1 The constraint on the pressure gradient.

upper boundary discretization points. In a such case it has been pointed out that the the functional to be minimized is complicated by the presence of other additional terms and penalization is required. The problem also becomes stiff. Nozzle results presented in the literature, e.g. [13] and [14], show a similar stiff behavior even if the optimization problem solved is simpler, as the outlet geometry is fixed.
In the numerical test, a diffuser is studied where the pressure distribution at the wall is optimized for minimal axial deviation at the outlet. The maximum attainable pressure gradient is constrained in order to avoid premature flow detachment. In the usual adjoint optimal shape design formulation, flow constraints are accounted for either through additional Lagrange multipliers, which means there are additional PDEs to be solved, or by a penalization in the functional. In the following, the design problem of an air-intake is formulated and the gradient is derived in detail for this application which is solved using an inverse problem governed by the Navier-Stokes equations [12] and adjoint optimization.

2 The Mathematical Model

Let us cConsider a two-dimensional diffuser with total pressure, total temperature and flow angles imposed at the inlet; pressure is given at the outlet. The walls of the diffuser should be designed so that the flow at the outlet has minimal axial deviation, and so that the diffuser causes a given pressure rise with a constraint on the maximum wall pressure gradient. The constraint imposed qualitatively reflects the Stratford [17] semi-empirical separation criterium for decelerated turbulent boundary layers

$$c_p \sqrt{s \frac{d\, c_p}{d\, s}} = C s^{\frac{1}{10}} \tag{1}$$

where C is a constant function of the Reynolds number per unit length, s the wall curvilinear coordinate and c_p the pressure coefficient. We take $s \approx x$ and $c_p = 2\,\dfrac{p - p_{in}}{\rho U_{in}^2}$, where

in refers to the inlet section. As the dynamic pressure is approximately equal to the difference between the total pressure p^0 and the inlet pressure p_{in}, we obtain

$$\frac{d\,p}{d\,x} \approx \frac{d\,c_p}{d\,x}(p^0 - p_{in})$$

The maximum allowed pressure gradient at the wall $\Pi(x)$ is displayed in fig. 1. In addition, the gradient is also required to be positive. This simple problem can be encountered in the design of wind tunnels diffusers, airbreathing engine intakes, or turbomachine casings.

2.1 Flow model and inverse problem solution method

Compressible viscous flows are governed by the Navier-Stokes equations. This set of equations may be written in a compact integral conservative form as:

$$\frac{\partial}{\partial t}\int_{\mathcal{V}} \boldsymbol{U}\, d\mathcal{V} + \int_{\mathcal{S}} (\boldsymbol{H}_I - \boldsymbol{H}_V)\cdot \boldsymbol{n}\, d\mathcal{S} = 0 \tag{2}$$

where $\mathcal{V}$ represents an arbitrary volume enclosed in a surface $\mathcal{S}$. System (2) is reduced to non-dimensional form with the help of the following reference values: L for length, ρ_∞ for density, T_∞ for temperature, $\sqrt{RT_\infty}$ for velocity, RT_∞ for energy per unit mass and μ_∞ for viscosity. In particular, $\boldsymbol{U}$ is the vector of conservative variables while tensors $\boldsymbol{H}_I$, $\boldsymbol{H}_V$ contain the inviscid and the viscous fluxes, respectively.

$$\begin{aligned} \boldsymbol{U} &= \{\rho, \rho\,\boldsymbol{q}, E\}^T \\ \boldsymbol{H}_I &= \{\rho\,\boldsymbol{q}, p\,\boldsymbol{I} + \rho\,\boldsymbol{q}\otimes\boldsymbol{q}, (E+p)\,\boldsymbol{q}\}^T = \boldsymbol{F}\,\boldsymbol{i} + \boldsymbol{G}\,\boldsymbol{j} \\ \boldsymbol{H}_V &= \frac{\sqrt{\gamma M_\infty}}{\mathrm{Re}_\infty}\{0, \boldsymbol{\tau}, \kappa\nabla T + \boldsymbol{\tau}\cdot\boldsymbol{q}\}^T \end{aligned} \tag{3}$$

The viscosity μ is computed via Sutherland's law and the perfect gas relationship $p = \rho T$ completes the set of equations. The diffuser geometry is unknown, but it is obtained imposing a given pressure distribution $p = p_e(s)$ on the solid boundaries, as opposed to direct problems where the geometry is known and the no-through-flow condition applies on the diffuser walls. In this sense, for sake of conciseness, we can write the steady state of eq. 2 as $\boldsymbol{E}(\boldsymbol{U}, p_e) = 0$. This solution method is based on the ideas presented in [18] and [12]. The diffuser walls can be considered as deformable and impermeable surfaces fastened to the diffuser inlet section, that move under the effect of the imposed pressure. An initial wall configuration is guessed. The resulting transient is described by integrating the equations that govern the time dependent flow motion. There results to be two differences compared to a usual flow solver. The first is that the no-through-flow boundary condition at the wall is replaced by a condition of given pressure, the second is that the set of equations that must solved is increased by one, that is the kinematic equation governing the surface motion. This equation is obtained by imposing that the speed of the moving wall is locally equal to the normal flow velocity. Therefore, in terms of computational cost, the inverse problem solution is equivalent to a direct solution. At the end of the flow transient, the wall assume the shape which induces on it the prescribed pressure distribution. System (2) is solved using a finite volume technique [12] on time-varying computational domains. The integration in time is carried out according to a Godunov scheme with Flux Difference Splitting (FDS). The fluxes $\boldsymbol{H}_I$ are evaluated by solving the Riemann's problems pertinent to the discontinuities that take place at the interfaces of the cells. Second order of accuracy is achieved by an Essentially Non-Oscillatory (ENO) scheme [21]. The velocity and temperature gradients that are required to evaluate the viscous fluxes $\boldsymbol{H}_V$ are computed using a integral technique that utilizes central differences and applies Gauss's Theorem.

2.2 Variational formulation and adjoint equations

The wall pressure $p_e(s)$ that should be imposed on the diffuser walls which minimize the functional

$$D[p_e(s)] = \frac{1}{2}\int_{out}\left(\frac{v}{u}\right)^2 dy \tag{4}$$

has to be determined. In order to solve such a constrained extremum problem the following Lagrangian function is introduced:

$$\mathcal{L}(\boldsymbol{U}, p_e, \Lambda) = D + \int_\Omega {}^t\boldsymbol{\Lambda}\ \boldsymbol{E}(\boldsymbol{U}, p_e)\, d\Omega \tag{5}$$

where ${}^t\Lambda(x,y) = \{\lambda_1, \lambda_2, \lambda_3, \lambda_4\}$ are Lagrange multipliers. This approach allows one to treat the problem as unconstrained. A stationary configuration is found when the variation of $\mathcal{L}$ with respect to all its arguments, which are now considered independent functions, is 0. Computing $\delta\mathcal{L}$ as

$$\delta\mathcal{L} = \delta\mathcal{L}_U + \delta\mathcal{L}_{p_e} + \delta\mathcal{L}_\Lambda \tag{6}$$

with

$$\begin{aligned} \delta\mathcal{L}_U = \delta D_U &+ \int_\Sigma {}^t\boldsymbol{\Lambda}\,(\boldsymbol{F}_U\ n_x + \boldsymbol{G}_U\ n_y)\delta\,\boldsymbol{U}\ d\sigma \\ &- \int_\Omega ({}^t\boldsymbol{\Lambda}_x\ \boldsymbol{F}_U + {}^t\boldsymbol{\Lambda}_y\,\boldsymbol{G}_U)\delta\,\boldsymbol{U}\ d\Omega \end{aligned} \tag{7}$$

where Σ is the boundary of flow field Ω, and $\boldsymbol{F}_U$, $\boldsymbol{G}_U$ are the fluxes Jacobian matrices. All the contributions to $\delta\mathcal{L}$ must be 0 at the maximum. Hence, to find a stationary point, we enforce

$$\delta\mathcal{L}_U = 0 \qquad \delta\mathcal{L}_\Lambda = 0$$

In general this results in $\delta\mathcal{L}_{p_e} \neq 0$. To reach the minimum, we take δp_e so that $\delta\mathcal{L} = \delta\mathcal{L}_{p_e} < 0$. It should be noted that the variations of $\mathcal{L}$ with respect to the Lagrange multipliers Λ, simply yield the flow equations.

From the condition $\delta\mathcal{L}_U = 0$, the adjoint of the Euler equations and its boundary conditions are obtained:

$${}^t\boldsymbol{\Lambda}_x\ \boldsymbol{F}_U + {}^t\boldsymbol{\Lambda}_y\,\boldsymbol{G}_U = 0 \text{ in } \Omega \tag{8}$$

and

$$\left[\frac{v}{u}\frac{\partial}{\partial \boldsymbol{U}}\left(\frac{v}{u}\right)\, h(\Sigma) + {}^t\boldsymbol{\Lambda}\,(\boldsymbol{F}_U\; n_x + \boldsymbol{G}_U\; n_y)\right]\delta\,\boldsymbol{U} = 0 \quad (9)$$

on Σ. $h(\Sigma)$ is 1 at the outlet and 0 elsewhere. The wall adjoint boundary condition is peculiar to this problem and is hereafter considered.

At the moving wall, $\delta\,\boldsymbol{U}$ can be written in terms of only 3 independent variations, as $\delta p_e(s) = 0$. We take $\delta(\rho e)$ as the dependent variation and eq. 9 becomes

$$\left\{\begin{array}{ccc} 0 & n_x & n_y \\ 0 & un_x & un_y \\ 0 & vn_x & vn_y \\ 0 & n_x(e+p)/\rho & n_y(e+p)/\rho \end{array}\right\}\left\{\begin{array}{c}\delta\rho\\ \delta(\rho u)\\ \delta(\rho v)\end{array}\right\} = 0 \quad (10)$$

where we take into account that the wall velocity is 0 when the inverse problem solution is attained. Equation 9, then, translates into the single condition

$$\lambda_1 + u\,\lambda_2 + v\,\lambda_3 + \frac{e+p}{\rho}\lambda_4 = 0 \quad (11)$$

that has to be satisfied at the wall. The functional gradient is

$$\delta D_{p_e} = \int_{wall} (\lambda_2\, n_x + \lambda_3\, n_y)\,\delta p\, ds \quad (12)$$

2.3 Pressure parameterization

The diffuser has imposed inlet p_{in} and outlet p_{out} wall pressures. The distributed control is the wall pressure gradient. On the discrete level, the pressure is recovered as

$$p_e(x_i) = p_{in} + \sum_{j=2}^{i} m(x_j)\Delta x_j, \qquad m(x_j) = \left(\frac{dp_e}{dx}\right)_j \quad (13)$$

with the constraint

$$\sum_{j=2}^{N} m(x_j)\,\Delta x_j = p_{out} - p_{in} \quad (14)$$

to match the exit pressure. Δx_j is the grid size in the x-direction and N is the number of computational points in the x-direction. We have $N-1$ control parameters represented by the pressure gradient at the discretization points.

Let us consider eq. 12 and discretize it as

$$\delta D_{p_e} = \sum_{j=2}^{N} \chi_j\,\delta p_j \quad (15)$$

with $\chi_j = [(\lambda_2\, n_x + \lambda_3\, n_y)\,\Delta s]_j$ and $\delta p_j = \sum_{i=2}^{j} \delta m(x_i)\Delta x_i$.

We have

$$\delta D_{p_e} = \sum_{i=2}^{N} \delta m(x_i)\Delta x_i \sum_{j=i}^{N} \chi_j \quad (16)$$

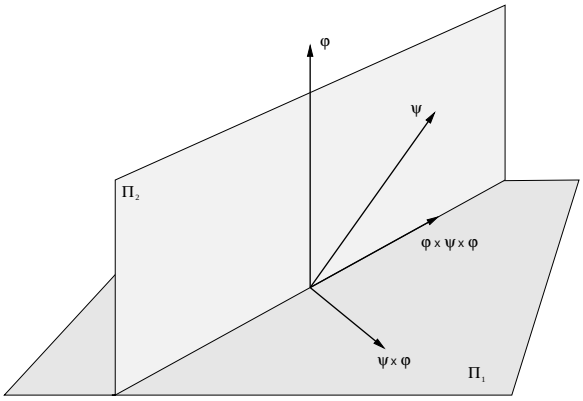


Fig. 2 Projection of the gradient in the constraint space.

Take $\psi_i = \Delta x_i \sum_{j=i}^{N} \chi_j$. If there were no constraints on the pressure gradient, we could simply set $\delta m(x_i) = -\varrho\,\psi_i$ to obtain $\delta D < 0$. Yet, in view of eq. 14, we also obtain

$$\sum_{i=2}^{N} \varphi_i\,\delta m(x_i)\,\Delta x_i = 0 \quad (17)$$

If $0 \le m(x_i) \le \Pi(x_i)$, then $\varphi_i = 1$, otherwise we take $\delta m(x_i) = 0$ and $\varphi_i = 0$. By projecting the gradient on the plane tangent to the constraints, see fig. 2, we have $\delta D < 0$ by taking

$$\delta m(x_i) = \varrho\vartheta(x_i) = \varrho[\{\phi_i\} \times \{\psi_i\} \times \{\phi_i\}] \quad (18)$$

The solution of the optimization problem is achieved by initializing the coefficients $m(x_j)$, computing the corresponding wall geometry using the inverse problem, solving the adjoint equations and updating the coefficients $m(x_j)$, according to the projected gradient, until the minimum is reached. This method, namely the gradient method, has a very slow convergence rate. Better convergence rates are obtained with the conjugate gradient method [22], in which the correction to $\delta m(x_i)$ at the iterate k is

$$(\delta m)^k = \varrho\,\left[\vartheta^k - \beta^{k-1}(\delta m)^{k-1}\right]$$

with

$$\beta^{k-1} = \frac{\displaystyle\int_{\Omega_b} [\vartheta^k - \vartheta^{k-1}]\,\vartheta^k d\Omega}{\displaystyle\int_{\Omega_b} [\vartheta^{k-1}]^2\, d\Omega}$$

2.4 Adjoint equation numerical solution

The numerical solution of the adjoint equations is obtained by using a first-order time-dependent technique based on a finite volume discretization. The solver computes the fluxes at cell interfaces using a flux-vector splitting technique. In a similar way, the boundary conditions are imposed on the numerical fluxes at the computational field edges.

Let us consider the adjoint equations. If a time derivative ${}^t\boldsymbol{\Lambda}_\tau$ is added to eq. 8, we are led to the hyperbolic system

$${}^t\boldsymbol{\Lambda}_\tau - {}^t\boldsymbol{\Lambda}_x \, \boldsymbol{F_U} - {}^t\boldsymbol{\Lambda}_y \, \boldsymbol{G_U} = 0 \tag{19}$$

This system is linear, because $\boldsymbol{F_U}$, $\boldsymbol{G_U}$, only depend on x and y, and its characteristics are the same as those of the flow problem, but with opposite speeds.

In order to take advantage of a finite volume formulation which is similar to that used for the flow equations we set

$${}^t\boldsymbol{\Lambda}_x \, \boldsymbol{F_U} = ({}^t\boldsymbol{\Lambda} \, \boldsymbol{F_U})_x - {}^t\boldsymbol{\Lambda}(\boldsymbol{F_U})_x \tag{20}$$

$${}^t\boldsymbol{\Lambda}_y \, \boldsymbol{G_U} = ({}^t\boldsymbol{\Lambda} \, \boldsymbol{G_U})_y - {}^t\boldsymbol{\Lambda}(\boldsymbol{G_U})_y \tag{21}$$

then, substituting by in eq. 19, we have

$${}^t\boldsymbol{\Lambda}_\tau - [{}^t\boldsymbol{\Lambda} \, \boldsymbol{F}_U]_x - [{}^t\boldsymbol{\Lambda} \, \boldsymbol{G}_U]_y + {}^t\boldsymbol{\Lambda}[(\boldsymbol{F}_U)_x + (\boldsymbol{G}_U)_y] = 0 \tag{22}$$

Considering an elementary volume of integration ω with surface σ, we rewrite eq. 22 in conservation form and apply the Gauss theorem to obtain

$$\frac{\partial}{\partial \tau} \int_\omega {}^t\boldsymbol{\Lambda} \, d\Omega - \int_\sigma {}^t\boldsymbol{\Lambda} \, \boldsymbol{C} \, d\sigma + {}^t\boldsymbol{\Lambda} \int_\sigma \boldsymbol{C} \, d\sigma = 0 \tag{23}$$

with $\boldsymbol{C} = \boldsymbol{F_U} \, n_x + \boldsymbol{G_U} \, n_y$. In the previous formula we considered Λ piecewise constant over the discretization volume. A characteristic-based approach is used to evaluate the convective fluxes at the cell interfaces. The total flux across the interface (int) is evaluated as the sum of two contributions which arise from the left (l) and right (r) sides of the interface, according to the wave-propagating nature of the hyperbolic system

$$({}^t\boldsymbol{\Lambda} \, \boldsymbol{C})_{int} = ({}^t\boldsymbol{\Lambda}^+ \, \boldsymbol{C}^+)_l + ({}^t\boldsymbol{\Lambda}^- \, \boldsymbol{C}^-)_r \tag{24}$$

where

$$\boldsymbol{C}^+ = \boldsymbol{L} \, \boldsymbol{D}^+ \, \boldsymbol{R}, \qquad \boldsymbol{C}^- = \boldsymbol{L} \, \boldsymbol{D}^- \, \boldsymbol{R} \tag{25}$$

and $\boldsymbol{D}^+ + \boldsymbol{D}^- = \boldsymbol{D}$. The matrix D is

$$\boldsymbol{D} = \begin{pmatrix} V_n & 0 & 0 & 0 & 0 \\ 0 & V_n & 0 & 0 & 0 \\ 0 & 0 & V_n & 0 & 0 \\ 0 & 0 & 0 & V_n - a & 0 \\ 0 & 0 & 0 & 0 & V_n + a \end{pmatrix} \tag{26}$$

The matrices $\boldsymbol{D}^+$and $\boldsymbol{D}^-$ are diagonal as well and they consist of the positive and negative eigenvalues of C respectively.

3 Results

In the following section we present the results for the diffuser test case. The diffuser is discretized over a 50×52 stretched grid. The inlet pressure is $p_{in} = 0.656$, the outlet pressure $p_{out} = 0.784$. The imposed flow angle at the inlet varies from zero, on the axis, to 5 degrees, at the upper wall. We are looking for the diffuser geometry that best approximates a zero flow angle at the outlet without flow separations. The latter requirement is here considered as equivalent to satisfy the constraint on the pressure gradient represented in fig. 1 . As already explained, the control is the pressure gradient at each computational point lying on the upper wall. The number of design variables is therefore equal to the number of grid cells in the x direction $(49 in this case)$. If a usual shape optimization method were to be applied in this case, we would obtain as many additional adjoint partial differential equations as the number of design variables, i.e. 49. The initial wall pressure distribution is a parabolic profile that satisfies the constraint on the pressure gradient in fig. 1. The final geometry of the diffuser is depicted in figures 3- 4 together with the Mach number isolines and

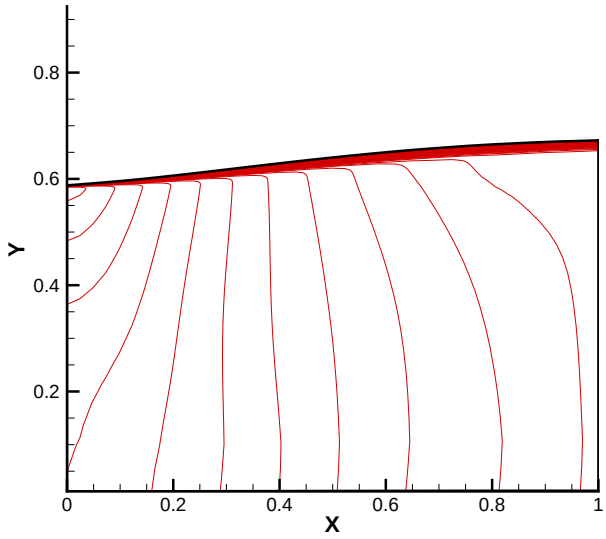


Fig. 3 Geometry and Mach number contours after the optimization process.

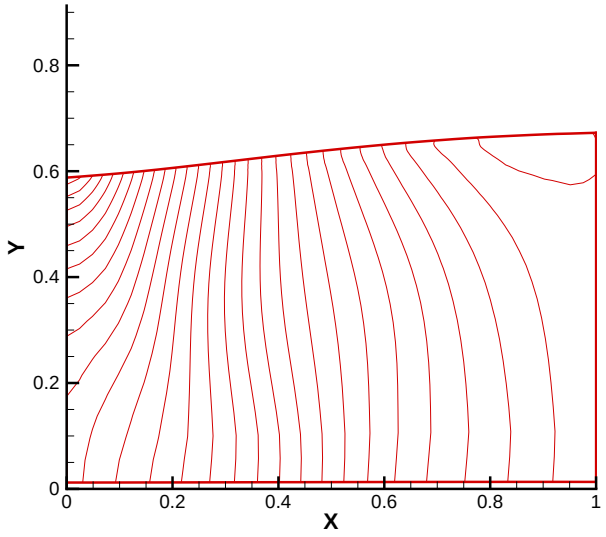


Fig. 4 Geometry and pressure field after the optimization process.

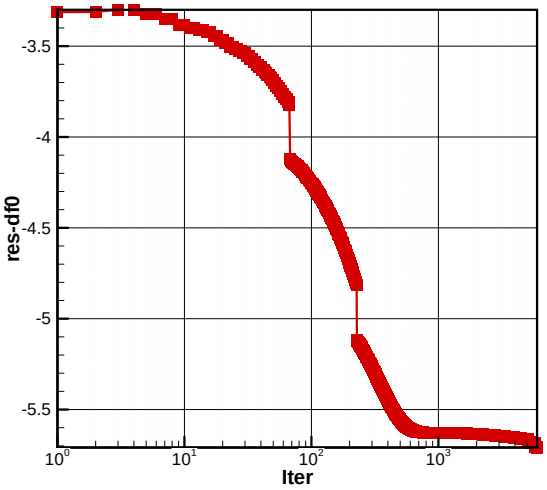


Fig. 5 Gradient residual versus optimization step.

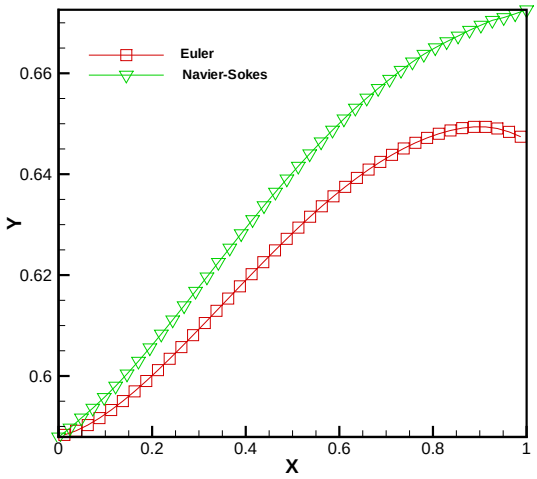


Fig. 7 Comparison between the optimal wall geometry obtained by the procedure in [11], based on the Euler equations, and the present viscous solution..

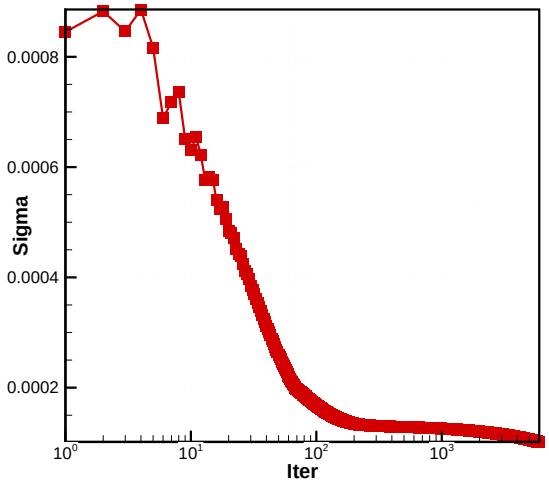


Fig. 6 Flow alignment versus optimization step.

pressure field, respectively . In figure 7 the wall geometry obtained by using the present procedure is compared with the solution obtained by applying the previously developed inviscid procedure proposed in [11].

The initial geometry is characterized by a non-zero flow angle $\sigma(y)_{out}$ at the exit. The l^2 norm of the gradient residuals is presented in figure 5. After 300 optimization steps the residual is decreased of more than two order of magnitude. The functional D decreases noticeably (see fig. 6). In about 100 optimization steps D is about a quarter of its initial value. Nevertheless, in order to show convergence and consistency of the approach presented in the previous sections, we made sure that the gradient becomes negligible. We have therefore pursued the number of optimization steps far beyond the point where the functional has a substantial decrease: we reached O(10000) optimization steps. From the point of view of applications, only 50-100 optimization steps would be enough to reach a substantial decrease in the functionals. After the first few optimization steps, the corrections to the flow as well as to the adjoint solution also become so small that only a few relaxation steps in the respective solvers are needed for convergence. In fact the solution of the first viscous inverse problem and adjoint procedure required about the half of the computational time. The case presented requires less that one hour of CPU time on a Dual-core Dual Opteron workstation to run about 6000 optimization steps.

4 Conclusions

The feasibility of a tool for the optimal design of diffusers and air intakes has been presented. The approach couples a viscous inverse technique with an adjoint optimization method. In general, constraints on the flow variables can be easily included in the optimization problem without the need of penalization techniques, thus avoiding the deterioration of the convergence rates and, eventually, stiffness. The relevant number of control variables involved in and the related computational cost in evaluating the gradient has shown the effectiveness of using adjoint methods. The approach to inverse problem solution, which essentially tracks a unsteady flow surface until a steady configuration is reached, could be further enhanced using over-relaxing techniques.

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References

1. W. Mangler, Die Berechnung eines Tragflugelprofiles mit vorgesc hriebener Druckverteilung. Jahrb. Deutsch. Luftfahrtforschung 1, 46-53, 1938.
2. J.M. Lighthill, A new method of two-dimensional aerodynamic design, Aeronaut. Res. Counc. Repts. and Mem. 2112, 1945.
3. A.M. Elizarov, N.B. Il'inskiy, A.V. Potashev, Mathematical methods of airfoil design, Akademie Verlag, Berlin, 1997.
4. L. Polito, Un metodo esatto per il progetto di profili alari in corrente incompressibile aventi un prestabilito andamento della velocià sul contorno, Università di Pisa, Rapporto Istituto di Aeronautica 42, 1974.
5. F. Bauer, P. Garabedian and D. Korn, Supercritical Wing Sections, Springer Verlag, Berlin, 1972.
6. G. Volpe, Geometric and surface pressure restrictions in airfoil design, AGARD-R-780, 1990.
7. O. Pironneau, On optimum design in fluid mechanics, JFM 59, 117-128, 1972.
8. A. Jameson, Aerodynamic design via control theory, ICASE Rep. 88-64, and also in J. Sci. Comp. 3, 233-260, 1988.
9. A. Jameson, Optimum aerodynamic design using control theory, CFD Review, 495-528, 1995.
10. A. Jameson, L. Martinelli, N.A. Pierce. Optimum aerodynamic design using the Navier-Stokes equations. TCFD 10, 213-237, 1998.
11. A. Iollo, M. Ferlauto, and L. Zannetti. An aerodynamic Optimization Method Based on the Inverse Problem Adjoint Equations. *J. Comp. Phys.*, 173:87–115, 2001.
12. M. Ferlauto, R. Marsilio. A Viscous Inverse Method for Aerodynamic Design, *Computers & Fluids*, 35 304–325,
13. F. Beux, A. Dervieux. Exact-gradient shape optimization of a 2-D Euler flow. Finite Elements in Analysis and Design 12, 281-302, 1992.
14. A. Iollo, M. D. Salas, Contribution to the optimal shape design of 2D interna flows with embedded shocks, JCP 125, 124-134, 1996.
15. R.F. van den Dam, J.A. van Egmond, J.W. Sloof, Optimization of target pressure distributions, AGARD-R-780, 1990.
16. J. Elliot, J. Peraire. Constrained, multipoint shape optimization for complex 3D configurations. The Aeronautical Journal, August/September, 365-376, 1998.
17. B.S. Stratford, The prediction of separation of the turbulent boundary layer, JFM 5, 1-35, 1959.
18. L. Zannetti, A time dependent method to solve the inverse problem for internal flows. AIAA J. 18, 754-758, 1980.
19. L. Zannetti, F. Larocca, Inverse methods for 3D internal flows, AGARD-R-780, 1990.
20. M. Pandolfi, A contribution to the numerical prediction of unsteady flows, AIAA J. 22, 602-610, 1984.
21. A. Harten, B. Engquist, S. Osher, Uniformly High Order Accurate Essentially Non-oscillatory Schemes, III, JCP 71, 231-303, 1987.
22. R. Fletcher, Practical methods of Optimization, John Wiley & Sons, 1980.